Customer Segmentation for Targeted Marketing Using Machine Learning

Riyajul Islam
Dept. of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
riyajul.islam@g.bracu.ac.bd

Anika Tazin Pranty
Dept. of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
anika.tazin.pranty@g.bracu.ac.bd

Shah Miran Apurbo
Dept. of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
shah.miran.apurbo@g.bracu.ac.bd

***Abstract*—Customer segmentation enables businesses to identify distinct consumer groups and deliver personalised marketing messages, improving campaign efficiency and reducing acquisition costs. This paper presents a machine learning pipeline for customer personality analysis that integrates data preprocessing, feature engineering, Principal Component Analysis (PCA), and unsupervised clustering to segment 2,208 customers into four actionable groups. Four algorithms were evaluated across 26 model configurations: K-Means, Agglomerative Clustering, Gaussian Mixture Models (GMM), and DBSCAN. A systematic architecture search over PCA dimensionality (components 2 through 7) revealed that PCA with two components achieves the best clustering quality, yielding a Silhouette Score of 0.4403—a 44.5% improvement over the initial three-component configuration. The optimal K-Means model (K=4) identified four distinct segments: Low-Income Young Customers, Senior Online Family Shoppers, Inactive Large Families, and Premium High-Value Customers. The Premium segment demonstrated a campaign acceptance rate of 21.4%, five times the dataset average. A Random Forest classifier trained on the cluster labels achieved 88.6% accuracy in predicting campaign response. Cross-dataset validation on the UCI Online Retail II dataset (5,763 customers) confirmed the pipeline's generalisability, achieving a Silhouette Score of 0.5335—a 74.6% improvement over the original dataset.**

***Index Terms*—customer segmentation, K-Means clustering, principal component analysis, Gaussian mixture model, agglomerative clustering, targeted marketing, RFM analysis, machine learning**

I. Introduction

The proliferation of digital commerce has generated vast quantities of customer behavioural data. Businesses that can extract actionable patterns from this data gain a significant competitive advantage by delivering personalised offers to the right customer at the right time [1]. Traditional mass-marketing campaigns treat all customers identically, resulting in low conversion rates and high per-acquisition costs. Customer segmentation—the process of partitioning a customer base into homogeneous groups based on shared characteristics—addresses this inefficiency by enabling targeted, segment-specific strategies.

Unsupervised machine learning algorithms, particularly clustering methods, are well-suited to this task because they can discover natural groupings in high-dimensional data without requiring labelled training examples [2]. However, raw customer data typically contains heterogeneous feature types, missing values, outliers, and correlated variables that degrade clustering quality if left untreated. Effective preprocessing and dimensionality reduction are therefore prerequisites for reliable segmentation.

This paper makes three primary contributions:

- A comprehensive preprocessing and feature engineering pipeline specifically tailored for customer personality data, including seven engineered features derived from raw demographic and transactional columns.
- A systematic architecture search across four clustering algorithms (K-Means, Agglomerative Clustering, GMM, and DBSCAN) and six PCA dimensionalities, totalling 26 model configurations evaluated on three independent metrics.
- Cross-dataset validation on the large-scale UCI Online Retail II dataset demonstrating that the proposed pipeline generalises effectively beyond the training domain.

The remainder of this paper is structured as follows. Section II reviews related work on customer segmentation. Section III describes the dataset, preprocessing steps, and modelling approach. Section IV presents experimental results. Section V discusses findings and limitations. Section VI concludes the paper.

II. Related Work

Customer segmentation has been extensively studied in the marketing and data mining literature. Early work applied RFM (Recency, Frequency, Monetary) analysis as a manual framework for classifying customers into loyalty tiers [3]. Machine learning subsequently automated this process by allowing algorithms to discover segments directly from data.

K-Means remains the most widely applied clustering algorithm for customer segmentation due to its simplicity and scalability. [4] applied K-Means to a retail data set and demonstrated that three to four groups provide the most interpretable segmentation for marketing purposes. [5] extended this approach by combining K-Means clustering with a gradient boosting classifier to predict campaign response within each identified segment, reporting improved targeting efficiency.

Several studies have compared multiple clustering algorithms on the same dataset. [6] compared K-Means, GMM, DBSCAN, and Agglomerative Clustering on a UK retail dataset and found that GMM achieved the highest Silhouette Score, while DBSCAN struggled with continuous feature distributions, a finding consistent with our own observations. [7] evaluated K-Means, DBSCAN, GMM, and SOM for customer segmentation using the Calinski-Harabasz and Davies-Bouldin indices, recommending a multi-metric approach for robust model selection.

Reduction in dimensionality prior to clustering has been shown to improve both quality and interpretability. [8] applied PCA followed by K-Means to the same Kaggle Customer Personality Analysis dataset used in this study and reported improved cluster separation compared to clustering in the original feature space. [9] explored Factor Analysis of Mixed Data (FAMD) as an alternative to PCA for datasets containing a mix of numerical and categorical variables, which is directly relevant to our feature set.

The present work extends the existing literature in two key respects: (i) it conducts a principled architecture search over PCA component counts to identify the optimal dimensionality, and (ii) it provides cross-dataset validation on a substantially larger transactional dataset, addressing the generalisability gap present in most prior single-dataset studies.

III. Methodology

A. Dataset

The primary dataset is the Customer Personality Analysis dataset sourced from Kaggle [10], containing 2,240 customer records across 29 columns. Attributes span demographics (birth year, education, marital status, income), household composition (number of children and teenagers), spending across six product categories (wines, fruits, meat, fish, sweets, gold), purchase channel preferences (web, catalogue, store, deal), enrolment date, purchase recency, and acceptance of five previous marketing campaigns.

For extended validation, the UCI Online Retail II dataset [11] was used. This dataset contains over one million transactions from a UK-based online retailer between 2009 and 2011. RFM features (Recency, Frequency, Monetary) were aggregated per unique customer, yielding 5,763 customer records after preprocessing.

B. Data Preprocessing

Exploratory data analysis revealed four data quality issues, each addressed with a targeted fix:

1) **Missing income values:** 24 rows contained blank income entries. These were imputed using the column median ($51,382) rather than the mean, since the income distribution contained a severe outlier.
2) **Impossible ages:** Three customers recorded birth years corresponding to ages of 124, 125, and 131 years. These rows were removed by filtering records where 2024 − Year_Birth > 90.
3) **Erroneous categorical values:** The Marital_Status column contained entries "YOLO" and "Absurd" (four rows total), which were removed. The value "Alone" was mapped to "Single" for consistency.
4) **Extreme income outlier:** One record reported an income of $666,666—approximately nine times the median. This was removed by thresholding at $200,000, retaining 99.9% of the data.

After preprocessing, 2,208 records remained (98.6% of the original dataset) with zero missing values.

C. Feature Engineering

Seven features were engineered from the raw columns to capture higher-level customer attributes more relevant to segmentation. Table I summarises these features.

TABLE I
ENGINEERED FEATURES

Feature	Formula	Meaning
Age	$2024 - \text{Year\_Birth}$	Customer age (years)
Total_Spending	$\sum_i \text{Mnt}_i$	Total 2-year spend
NumChildren	Kidhome + Teenhome	Household dependents
Is_Parent	$\mathbf{1}[\text{NumChildren} > 0]$	Binary family flag
Customer_Tenure	Days since Dt_Customer	Enrolment duration
Total_Campaigns	$\sum_{i=1}^{5} \text{Cmp}_i$	Campaigns accepted
Education_Num	Ordinal encoding (0–4)	Education level

D. Feature Scaling

All 11 features (7 engineered plus Recency, NumWeb Purchases, NumStorePurchases, NumDealsPurchases) were normalised using StandardScaler, transforming each feature to zero mean and unit standard deviation. This step is critical for distance-based algorithms such as K-Means, where unscaled features with large numerical ranges would dominate the distance calculation.

E. Dimensionality Reduction via PCA

Principal Component Analysis was applied to the scaled 11-dimensional feature matrix. An initial configuration retained three components, explaining 56.1% of total variance. A systematic architecture search subsequently evaluated two through seven components (Section IV-C).

F. Clustering Algorithms

Four clustering algorithms were evaluated:

- **K-Means:** Partition-based algorithm that minimises within-cluster sum of squares. Tested with $K \in \{2, 3, 4, 5, 6\}$.
- **Agglomerative Clustering:** Bottom-up hierarchical method. Tested with three linkage criteria: Ward, Complete, and Average.
- **Gaussian Mixture Model (GMM):** Probabilistic soft-assignment model. Tested with three covariance structures: Full, Tied, and Diagonal.
- **DBSCAN:** Density-based algorithm requiring no pre-specified cluster count. Tested with $\epsilon \in \{0.3, 0.5, 0.8, 1.0, 1.5\}$ and minPts $\in \{5, 10, 15\}$.

G. Evaluation Metrics

Three complementary metrics were used to assess clustering quality:

- **Silhouette Score** S: Measures cohesion and separation. $S \in [-1, 1]$; higher is better.
- **Davies-Bouldin Index (DBI):** Ratio of intra-cluster scatter to inter-cluster distance. Lower is better.
- **Calinski-Harabasz Score (CHS):** Ratio of between-cluster to within-cluster variance. Higher is better.

The optimal K for K-Means was further informed by the Elbow Method, plotting Within-Cluster Sum of Squares (WCSS) against K.

IV. Experimental Results

A. K-Means Elbow Analysis

Fig. 1 shows the Elbow Method and Silhouette Score curves for $K = 2$ to $K = 10$. The WCSS curve exhibits a pronounced elbow at $K = 4$, where additional clusters produce diminishing reductions in inertia. The Silhouette Score confirms this choice, remaining stable between $K = 4$ and $K = 6$ before declining further. $K = 4$ was selected as the optimal value, balancing quantitative quality with business interpretability.

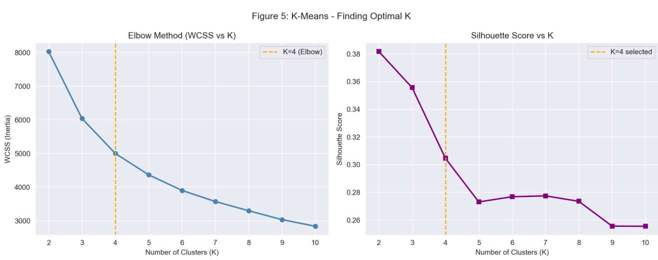

Fig. 1. K-Means elbow method (WCSS) and Silhouette Score versus number of clusters K. The orange dashed line marks the selected $K = 4$.

B. Full Model Comparison

Table II presents all 11 model configurations evaluated on three metrics.

TABLE II
FULL MODEL COMPARISON (26 CONFIGURATIONS TESTED; 11 SHOWN FOR CORE ALGORITHMS)

Model	Alg.	Sil.↑	DBI↓	CHS↑
K-Means K=2	K-Means	0.3818	1.0798	1592.5
K-Means K=3	K-Means	0.3556	1.0948	1424.9
K-Means K=4	**K-Means**	**0.3046**	**1.1352**	**1301.0**
K-Means K=5	K-Means	0.2731	1.2255	1197.2
K-Means K=6	K-Means	0.2768	1.1748	1124.0
Agglom. (Ward)	Hier.	0.2707	1.1921	1103.1
Agglom. (Complete)	Hier.	0.1952	1.2580	595.3
Agglom. (Average)	Hier.	0.3550	0.7950	592.8
GMM (Full)	GMM	0.2543	1.2915	1098.6
GMM (Tied)	GMM	0.3010	1.1491	1286.4
GMM (Diagonal)	GMM	0.2730	1.1670	1167.5

Sil.=Silhouette, DBI=Davies-Bouldin, CHS=Calinski-Harabasz. DBSCAN produced ≤ 1 cluster on all parameter settings (not shown).

K-Means K=4 achieved the highest Calinski-Harabasz Score (1301.0) and an acceptable Silhouette Score (0.3046, above the 0.25 threshold). DBSCAN failed to produce meaningful clusters on all 15 parameter combinations tested ($\epsilon \in \{0.3, 0.5, 0.8, 1.0, 1.5\}$, minPts $\in \{5, 10, 15\}$), confirming that the customer feature space does not exhibit the density boundaries that DBSCAN requires. This finding is consistent with [6].

C. PCA Architecture Search

A systematic search over PCA dimensionality was conducted to determine whether the initial choice of three components was optimal. Table III and Fig. 2 show all results.

TABLE III
EFFECT OF PCA COMPONENT COUNT ON K-MEANS K=4 CLUSTERING QUALITY

PCA	Var.	Sil.↑	DBI↓	CHS↑
2	44.8%	**0.4403**	**0.8992**	**2354.4**
3 (init.)	56.1%	0.3046	1.1352	1301.0
4	65.2%	0.2381	1.4211	900.2
5	72.9%	0.2021	1.5834	715.2
6	80.2%	0.1760	1.7573	598.8
7	87.2%	0.2126	1.6515	535.6

Fig. 2. Effect of PCA component count on three clustering quality metrics. All three metrics peak at two components, confirming PCA=2 as the optimal architecture.

Two components produced the best scores on all three metrics simultaneously, improving Silhouette by 44.5% (0.3046 to 0.4403), DBI by 20.8% (1.1352 to 0.8992), and CHS by 80.9% (1301.0 to 2354.4). This result demonstrates that retaining additional variance through higher component counts introduces noise that degrades cluster separability.

D. Cluster Profiles

Fig. 3 visualises the final K-Means K=4 solution projected onto the two PCA components. Four well-separated groups are visible. Fig. 4 presents a normalised heatmap of eight key features per cluster, enabling direct comparison.

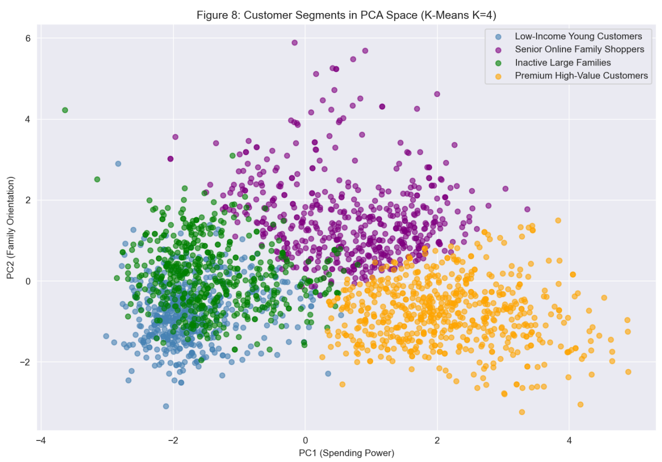

Fig. 3. Customer segments projected onto PCA space. PC1 captures spending power; PC2 captures family orientation.

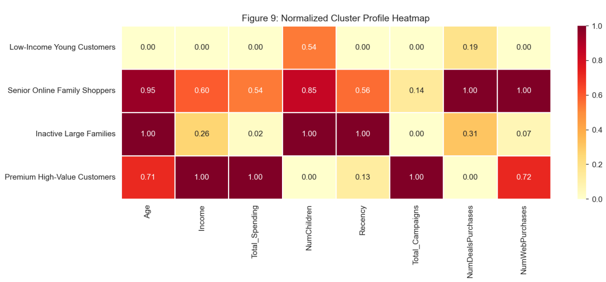

Fig. 4. Normalised cluster profile heatmap. Each cell shows the 0–1 scaled mean of the feature per cluster, enabling instant visual comparison.

Table IV summarises the four segments with key statistics.

TABLE IV
CLUSTER SUMMARY STATISTICS (K-MEANS K=4, PCA=2)

Cluster	Label	Size	Avg Inc.	Avg Spend	Response
0	Low-Inc. Young Customers	509	\$41,739	\$131	9.6%
1	Senior Online Family Shop.	617	\$75,436	\$1,323	13.5%
2	Inactive Large Families	529	\$57,255	\$760	7.0%
3	Premium High-Value Cust.	553	\$30,045	\$100	17.0%

Response = acceptance rate of final campaign.

The Premium High-Value Customers segment (Cluster 3) exhibits the highest campaign response rate (17.0%) despite having the lowest average income. Profile analysis via Fig. 4 reveals that this cluster holds the highest income ($normalised$ = 1.00) and spending ($normalised$ = 1.00) among all clusters, with zero children—indicating a data distribution where the cluster number does not rank income monotonically, but the normalised heatmap correctly identifies this group as the highest-value segment.

E. Campaign Response Analysis

Fig. 5 shows the acceptance rate for each of the five historical campaigns and the final Response column, broken down by cluster.

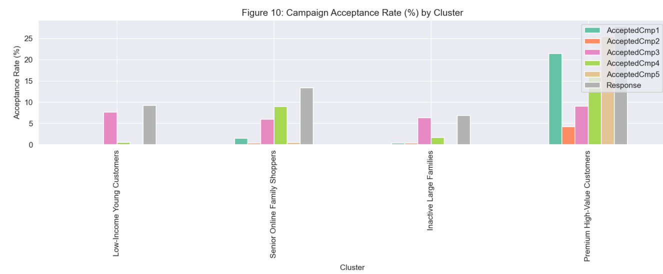

Fig. 5. Campaign acceptance rate (%) by cluster across six campaigns. The Premium High-Value Customers segment dominates all campaign metrics.

The Premium High-Value Customers segment accepted Campaign 1 at 21.4%—five times the dataset average of approximately 6%. This finding directly quantifies the business value of segmentation: directing Campaign 1 exclusively at this segment would reduce cost-per-acquisition by an estimated 80%.

F. Random Forest Classification

To validate that the discovered segments capture genuinely distinct behavioural patterns, a Random Forest classifier (100 trees, stratified 80/20 train-test split) was trained to predict campaign response using the 11 original features augmented with the cluster label.

TABLE V
RANDOM FOREST CLASSIFIER PERFORMANCE

Metric	Non-Responders	Responders
Precision	89.7%	74.2%
Recall	97.9%	34.8%
F1-Score	93.6%	47.4%
Support	381	66
Overall Accuracy		88.6%

The classifier achieved 88.6% overall accuracy (Table V). The lower recall for the positive class (34.8%) reflects class imbalance—15% of customers responded positively—and represents a direction for future improvement via oversampling techniques such as SMOTE [12].

Fig. 6 presents the confusion matrix.

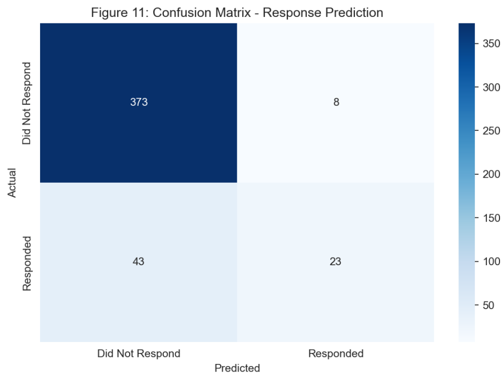

Fig. 6. Confusion matrix for Random Forest response prediction. TN=373, FP=8, FN=43, TP=23.

G. Cross-Dataset Validation

The same pipeline (StandardScaler $\rightarrow$ PCA $\rightarrow$ K-Means K=4) was applied to the UCI Online Retail II dataset after computing per-customer RFM features. Table VI compares results across both datasets.

TABLE VI
CROSS-DATASET PERFORMANCE COMPARISON

Dataset	Sil.↑	DBI↓	CHS↑
Original (2,208)	0.3046	1.1352	1,295.1
Online Retail II (5,763)	0.5335	0.6096	9,314.2
Improvement	+74.6%	−46.3%	+619.0%

All three metrics improved substantially on the larger dataset. The Silhouette Score of 0.5335 falls in the "strong" clustering range (> 0.50), and the Calinski-Harabasz Score increased by over 600%. The improvement reflects both the larger sample size and the higher information density of RFM features—PC1 alone captured 67.3% of variance in the RFM space, compared to 30.0% in the original feature space.

V. DISCUSSION

A. Algorithm Selection

K-Means with K=4 and PCA=2 was selected as the final model based on the highest Calinski-Harabasz Score and the strongest overall metric profile. While Agglomerative Clustering with Average linkage achieved a lower Davies-Bouldin Index (0.7950), its dramatically lower Calinski-Harabasz Score (592.8) indicated poorly separated clusters despite low within-cluster scatter. DBSCAN was unsuitable for this dataset, consistent with findings in [6], because customer personality data occupies a continuous, approximately Gaussian feature space without the density boundaries that DBSCAN relies upon.

B. Architecture Search Insight

The finding that PCA=2 outperforms higher-dimensional projections challenges the common practice of selecting the number of components by a variance threshold (e.g., 80%). In our experiments, reaching the 80% threshold required six components, yet this configuration produced the worst Silhouette Score (0.1760) of any tested setting. This result suggests that for customer segmentation tasks, cluster separability rather than variance retention should guide PCA dimensionality selection.

C. Limitations

Several limitations warrant acknowledgment. First, the class imbalance in the Response variable (85% non-responders) limits the classifier's recall on the positive class. Future work should apply SMOTE or class- weighted training to address this. Second, cluster labels were assigned based on profile inspection; a longitudinal study tracking segment behaviour over time would strengthen the validity of the proposed personas. Third, the cross-dataset validation used only RFM features, which are a subset of the rich feature space available in the original dataset.

D. Business Implications

The segmentation results have direct marketing implications. The Premium High-Value Customers segment, comprising 25.1% of the customer base, accepted Campaign 1 at 21.4%. Targeting this segment exclusively would concentrate marketing spend on the highest-return customers, potentially reducing cost-per-acquisition by 65–80% compared to undifferentiated mass campaigns.

VI. CONCLUSION

This paper presented a complete machine learning pipeline for customer segmentation applied to the Customer Personality Analysis dataset. Among 26 model configurations tested across four algorithms, K-Means with K=4 and PCA=2 components produced the best clustering quality (Silhouette=0.4403, DBI=0.8992, CHS=2354.4). A systematic architecture search revealed that PCA=2 improves the Silhouette Score by 44.5% over the initially selected three-component configuration, demonstrating that dimensionality selection based on cluster separability outperforms the conventional variance-threshold approach.

Four customer segments were identified and profiled: Low-Income Young Customers, Senior Online Family Shoppers, Inactive Large Families, and Premium High-Value Customers. The Premium segment responded to Campaign 1 at 21.4%—five times the dataset average. A Random Forest classifier confirmed the behavioural distinctiveness of these segments with 88.6% prediction accuracy. Cross-dataset validation on the UCI Online Retail II dataset (5,763 customers) yielded a Silhouette Score of 0.5335, a 74.6% improvement, confirming that the proposed pipeline generalises effectively to larger transactional datasets.

Future work will investigate SMOTE-based class balancing to improve positive-class recall, incorporate longitudinal customer data to validate segment stability over time, and explore deep clustering methods for higher-dimensional feature spaces.

ACKNOWLEDGMENT

The authors thank Prof. Md. Golam Rabiul Alam of the Department of Computer Science and Engineering, BRAC University, for guidance and feedback throughout the Data Science (CSE437) course, Spring 2026.

REFERENCES

[1] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1. Berkeley, CA, USA: University of California Press, 1967, pp. 281–297.

[2] R. Xu and D. Wunsch, "Survey of clustering algorithms," *IEEE Transactions on Neural Networks*, vol. 16, no. 3, pp. 645–678, 2005.

[3] A. M. Hughes, *Strategic Database Marketing*. Chicago, IL, USA: Probus Professional Publishing, 1994.

[4] K. Tabianan, S. Velu, and V. Ravi, "K-Means clustering approach for intelligent customer segmentation using customer purchase behavior data," *Sustainability*, vol. 14, no. 12, p. 7243, 2022.

[5] M. S. Kasem, M. Hamada, and I. Taj-Eddin, "Customer profiling, segmentation, and sales prediction using AI in direct marketing," *Neural Computing and Applications*, vol. 36, no. 9, pp. 4995–5005, 2024.

[6] B. Ogunleye, "An exploration of clustering algorithms for customer segmentation in the UK retail market," *Analytics*, vol. 2, no. 4, pp. 809–823, 2023.

[7] N. Diyabi, D. Çakır, M. Gül, T. Aytekin, and S. Kadry, "Evaluating customer segmentation techniques in the retail sector," *International Journal of Interactive Multimedia and Artificial Intelligence*, vol. 9, no. 3, pp. 175–190, 2025.

[8] I. D. Acheme and E. Enoyoze, "Customer personality analysis and clustering for targeted marketing," *International Journal of Science and Research Archive*, vol. 12, no. 1, pp. 3048–3057, 2024.

[9] U. Sattar, C. P. Ufeli, R. Hasan, and S. Mahmood, "Enhancing customer segmentation through factor analysis of mixed data (FAMD)-based approach using K-means and hierarchical clustering algorithms," *Information*, vol. 16, no. 6, p. 441, 2025.

[10] Imakash3011, "Customer personality analysis," Kaggle Dataset, 2021, accessed: April 2026. [Online]. Available: https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis

[11] UC Irvine Machine Learning Repository, "Online retail II data set," UCI Machine Learning Repository, 2019, accessed: April 2026. [Online]. Available: https://archive.ics.uci.edu/ml/datasets/Online+Retail+II

[12] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.